GREEN ALIBI

Testing Whether Fluorescence Catches Drought Stress Before the Eye Can See It

Executive Summary · DOI: 10.5281/zenodo.21762501 · Sakshi D. Maske

Project Overview

I built GREEN ALIBI to turn a physical hunch — that Solar-Induced Fluorescence should catch drought stress before NDVI does, because it tracks a drop in photosynthetic efficiency directly instead of waiting for a leaf's structure to visibly break down — into a testable, falsifiable claim rather than something policy just assumes. India's drought-declaration process, and the PMFBY crop-insurance payout mechanism riding on it, lean on NDVI because it's the established indicator, not because anyone has actually shown it's the fastest one; if SIF really has a physical head start, that gap translates directly into how quickly a stressed farmer gets paid. I framed the work around four research questions instead of a single hypothesis — does SIF actually lead NDVI, is that lag geographically consistent, how does it line up against the one official drought declaration I could verify, and is it large enough to matter for insurance timelines — because a genuine early-warning case needs more than one line of evidence behind it. The project started as a three-year pilot (2015, 2018, 2020), and I extended it to eight growing seasons specifically because three years wasn't enough to trust, re-running the full pipeline — acquisition, boundary definition, lag calculation — against the larger sample instead of just bolting new numbers onto old code. That expansion also surfaced this study's most important finding about itself: the original three-year sample turned out to be two-thirds drought years, far more drought-heavy than Marathwada's actual eight-year climate record, exactly the kind of thing a small sample tends to hide. What follows is the result of that harder test, H3 not holding up and 2018 stubbornly refusing to fit the pattern included.

The Question

India's official drought-declaration process, and the PMFBY crop-insurance payout mechanism that depends on it, rely substantially on NDVI — a reflectance signal that only changes once a plant's structure has already begun to visibly degrade. Solar-Induced Fluorescence (SIF) is grounded in a more direct physical process: a drop in photosynthetic efficiency, before outward greenness changes. Does that physical head-start actually show up as a measurable time lag on real satellite data — and does drought severity make that lag bigger, as hypothesized?

The Method

Eight districts of Marathwada, Maharashtra were tested across eight growing seasons (2015–2023, excluding 2021), using GOSIF v2 fluorescence data and cloud-screened MODIS NDVI. The study started with three years (2015, 2018, 2020); five more (2016, 2017, 2019, 2022, 2023) were added later specifically to fix the original small-sample limitation. The SIF-to-NDVI decline lag was calculated two independent ways — threshold-crossing (linear-interpolation at 5 seasonal decline thresholds) and time-lagged cross-correlation — with 2,000-replicate bootstrap confidence intervals quantifying precision. Rainfall-anomaly validation against a 20-year CHIRPS climatology found only 2 of the 8 years (2015, 2018) actually meet this study's own drought threshold — the other 5 added years all sit within roughly one standard deviation of normal.

The Finding

SIF's decline precedes NDVI's in 7 of the 8 years — confirmed in direction by an independent cross-correlation method, and by bootstrap resampling never favoring NDVI leading SIF in any single year (100% of replicates, every year). 2018 is a genuine, replicated exception: its threshold-crossing lag is marginally negative, its cross-correlation lag is exactly zero, and only 8.1% of its bootstrap replicates show a positive lag — three independent methods agreeing 2018 breaks the pattern. Drought severity still doesn't amplify the lag: the two drought years averaged a smaller lag (7.6 days) than the six normal years (15.0 days) — the same direction as the original 3-year study, now confirmed at more than double the sample size.

Year	Rainfall Anomaly	Threshold-Crossing Lag	Cross-Corr. Lag
2015 (drought)	−21.5%	16.3 days	4 days
2016 (normal)	+9.5%	23.8 days	2 days
2017 (normal)	−1.1%	21.6 days	27 days
2018 (drought)	−18.3%	−1.1 days	0 days
2019 (normal)	+13.3%	17.2 days	18 days
2020 (normal)	+29.1%	20.6 days	0 days
2022 (normal)	+36.8%	5.1 days	0 days
2023 (normal)	−3.7%	4.2 days	0 days

H1 (SIF leads NDVI) holds up under both methods in most years, with 2018 an honestly-disclosed exception across all of them. H3 (drought amplifies the lag) is reported plainly as not supported — a real non-result, confirmed at n = 8 years rather than just n = 3.

Validation & Robustness Checklist

✓ Two independent lag-estimation methods (threshold-crossing + cross-correlation)

✓ Bootstrap confidence intervals (2,000 replicates per year, all 8 years)

✓ Independent rainfall validation (CHIRPS, 20-year climatological baseline)

✓ Moran's I spatial-autocorrelation check (effective sample size disclosed, per year)

✓ Cross-referenced against the real government drought-declaration date (2018)

✓ Sample expanded from 3 to 8 years — a real, executed fix, not just a stated limitation

! H3 (drought amplifies lag) — honestly reported as not supported

! 2018 is a consistent exception to H1 across every method — reported, not hidden

! Exact year-to-year ranking flagged as method-sensitive, not robust

Honest Limitation

Only 2 of 8 years meet this study's own drought threshold, so the drought-vs-normal comparison stays a 2-versus-6 split, not a balanced one. Every lag value is a point estimate from just 14–18 post-peak satellite observations per year, so confidence intervals run wide — all 28 possible pairwise comparisons between years' intervals overlap, meaning the exact year-to-year ranking isn't statistically distinguishable from noise even at 8 years. The SIF-rainfall spatial correlation (Pearson $r = 0.567$, $p < 0.0001$, $n = 64$ district-years) is real but noticeably weaker than the $r = 0.837$ the smaller, drought-heavy 3-year sample originally produced, and a Moran's I diagnostic finds SIF's own spatial clustering significant in only 4 of 8 years, against rainfall's, significant in all 8.

Real-World Relevance

Comparing 2018's SIF and NDVI decline onset against Maharashtra's official drought declaration (31 October 2018) shows both satellite indicators beating the official declaration by roughly seven to eight weeks. After a district-boundary naming bug was found and fixed during the sample expansion, NDVI now crosses its threshold about five days before SIF in 2018 specifically — the reverse of the original finding, and in line with 2018 being this study's one exception throughout. For a rainfall-deficit-prone farming region, the bigger, more dependable opportunity is satellite-based monitoring of either kind arriving seven-plus weeks ahead of the current process, with SIF's own edge over NDVI a smaller, year-dependent bonus on top — not a guaranteed one.

GitHub: github.com/sakshimaske303-commits/GREEN_ALIBI | Live Dashboard: greenalibi-bzs2wvod5fflqh7dfe2cf4.streamlit.app | Zenodo DOI: 10.5281/zenodo.21762501

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